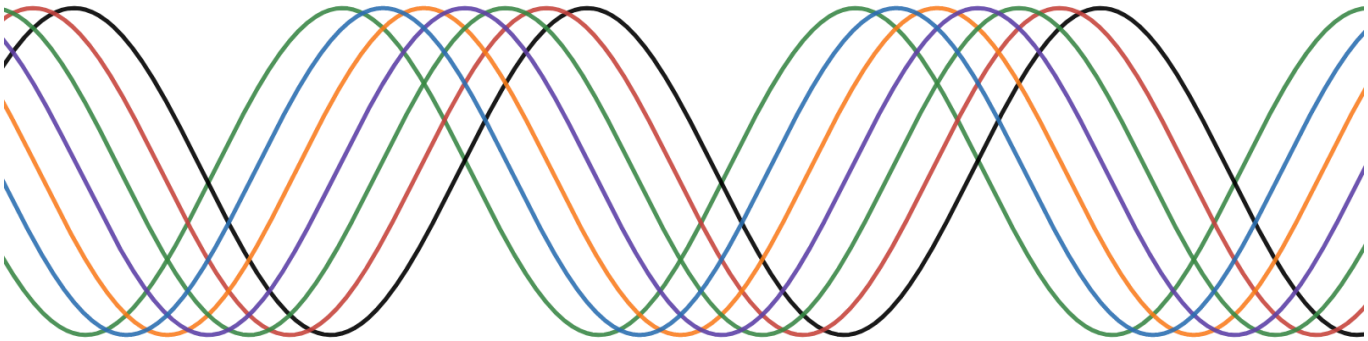
Math 112

PRECALCULUS

COURSE PACK



FALL 2026 / SPRING 2027

F1.A Changing Quantities

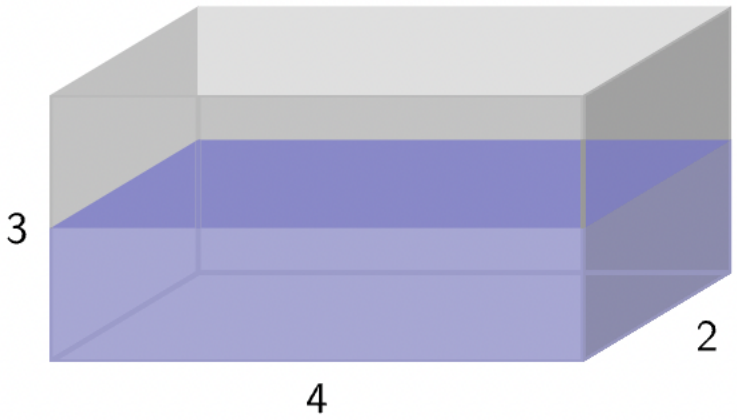
1. Suppose that a rectangular aquarium is being filled with water. The tank is 4 feet long by 2 feet wide by 3 feet high, and the hose that is filling the tank is delivering water at a rate of 0.5 cubic feet per minute.

   (a) What are some different quantities that are changing in this scenario?

Changing	**Staying the Same**

   (b) After 1 minute has elapsed, how much water is in the tank? At this moment, how deep is the water?

   (c) How long will it take for the tank to be completely full? Why?



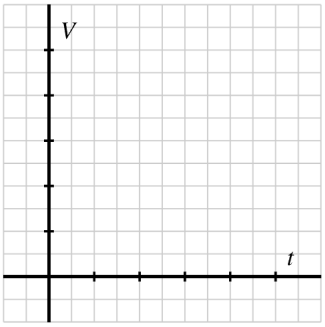
2. Consider a tank in the shape of an inverted circular cone (point down) where the tank's radius is 2 feet and its depth is 4 feet. Suppose that the tank is being filled with water that is entering at a constant rate of 0.75 cubic feet per minute.

   (a) Sketch a labeled picture of the tank, including a snapshot of there being water in the tank prior to the tank being completely full.

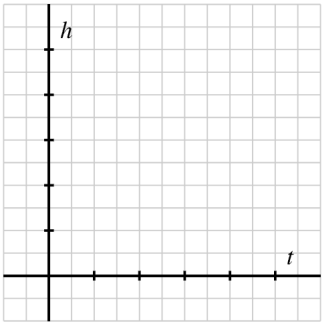
   (b) What are some quantities that are changing in this scenario? What are some quantities that are not changing?

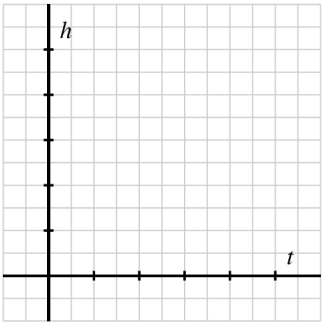
   (c) Fill in the following table of values to determine how much water, V, is in the tank at a given time in minutes, t, and thus generate a graph of the relationship between volume and time by plotting the data on the provided axes.

t	V
0	
1	
2	
3	
4	
5	



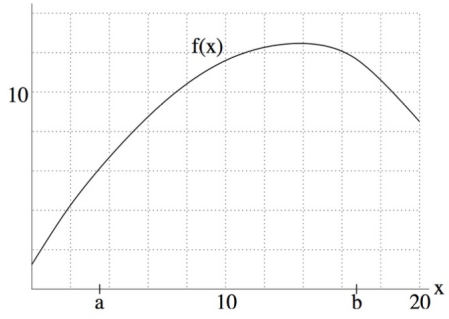
   (d) Finally, think about how the height of the water changes in tandem with time. Without attempting to determine specific values of h at particular values of t, how would you expect the data for the relationship between h and t to appear? Use the provided axes to sketch at least two possibilities; write at least one sentence to explain how you think the graph should appear.





F1.B Functions

1. Use the graph of the function f to find the following values. Estimate where appropiate.



(a) $f(10) =$ _____

(b) If $f(x) = 10$, then $x =$_____

(c) $f(a) =$_____

(d) $f(2a) =$_____

(e) $2f(a) =$_____

(f) $f(a + b) =$_____

Definitions: A **function** is a process that may be applied to a set of input values to produce a corresponding set of output values in such a way that the process produces one and only one output value for any single input value.

Let F be a function from A to B.

- The set A of possible inputs to F is called the _____ of F.
- The collection of actual outputs is called the _____ of F.

.

2. Identify the domain and range for the following functions.

(a) $f(x) = \dfrac{5}{x}$

(b) $f(x) = \dfrac{x}{5}$

(c) $f(x) = \sqrt{5x}$

3. Dolbear's model is given by $T = f(N) = 40 + 0.25N$ where T is the temperature in Fahrenheit, and N is the number of chirps per minute produced by snowy tree crickets.

   (a) What does $f(16) = 44$ mean in words? What are the units for 16? What are the units for $f(16)$?

   (b) If we hear snowy tree crickets chirping at a rate of 92 chirps per minute, what does Dolbear's model suggest should be the outside temperature?

   (c) If the outside temperature is 77°F, how many chirps per minute should we expect to hear?

4. Consider a spherical tank of radius 4m that is filling with water. Let V be the volume of water in the tank (in cubic meters) and h the depth of the water (in meters). V is a function of h according to the rule

$$V = f(h) = \frac{\pi}{3} h^2 (12 - h)$$

   (a) What values of h make sense to consider in the context of this function? What values of V make sense in the same context?

   (b) What is the domain of the function f in the context of the spherical tank? Why?

   (c) Determine and interpret (with appropriate units) the values $f(2)$, $f(4)$, and $f(8)$. What is important about the value of $f(8)$?

   (d) Compute $f(13)$. Does this value make sense? What went wrong?

   (e) Find a value of h for which $f(h) = 300$ cubic meters? How might a graph of the function help?

A1: Solving Equations and Inequalities

Learning Objectives

- Solve linear and quadratic equations
- Find solution sets for linear inequalities and inequalities with absolute values

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1. SOLVING EQUATIONS: Solve the following equations for x.

   (a) $3x - 4 = \frac{1}{2}(2 - x)$

   (b) $x(2x - 4) = 6$

   (c) $5x^2 - 3x - 20 = 0$

   (d) $(x - 5)^2 = 10$

   (e) $xyz = x(y + z) + y$

   (f) $\dfrac{x+1}{3} - \dfrac{x-1}{6} = \dfrac{1}{6}$

2. INEQUALITIES: Find the solution sets for the following equations or inequalities.

(a) $3x - 3 \leq 5$

(c) $\dfrac{x}{2} + \dfrac{3}{4} < \dfrac{5}{8}$

(b) $2(4 - x) > 6$

(d) $5 - 7x \leq 3x + 2$

3. ABSOLUTE VALUES: Find the solution sets for the following equations.

(a) $|2x + 3| = 11$

(c) $|3x - 1| = 2x + 7$

(b) $\left|\dfrac{x}{3} - 2\right| = 5$

(d) $\left|\frac{2x-1}{4}\right| = \frac{3}{2}$

A2: Factoring, Multiplying, and Rational Expressions

1. MULTIPLYING POLYNOMIALS: Multiply the following.

   (a) $3x(x+2)$

   (b) $(3x+4)(\frac{1}{2}x-2)$

   (c) $x^2(x-1)(x+2)$

   (d) $(x^2-3)^2$

   (e) $(2x-\sqrt{y})(2x+\sqrt{y})$

   (f) $5x(x-7)^2$

2. FACTORING: Factor each expression completely.

   (a) x^2-5x-6

   (b) $32t-t^2$

   (c) $36x^4-2x^2+16x^6$

   (d) $6x^2+7x-5$

   (e) $9x^2-16$

   (f) $4x^2-4xy+y^2$

3. SIMPLIFYING RATIONAL EXPRESSIONS: Identify fractions that can be simplified, or state that no simplification is possible.

(a) $\dfrac{3x^2}{9x}$

(b) $\dfrac{3x+1}{3x}$

(c) $\dfrac{2(t+3)}{4t(3+t)}$

(d) $\dfrac{3y^3}{3y-6} \cdot \dfrac{y-2}{y}$

(e) $\dfrac{3}{t-1} + \dfrac{t}{5}$

(f) $\dfrac{\frac{3}{x} - \frac{x}{3}}{\frac{1}{x} + \frac{1}{3}}$

A2.B Algebra Review

1. True or false and explain. Your explanation should give compelling evidence that the equation is true for all cases or false. Consider using algebra, graphs, and/or counter-examples to support your conclusion. Explain your reasoning. (Assume $a \neq 0$).

   a. $\dfrac{1}{a^2+2} = \dfrac{1}{a^2} + \dfrac{1}{2}$ TRUE FALSE

   b. $\dfrac{a-b}{a} = \dfrac{b-a}{-a}$ TRUE FALSE

2. Find the set of values of x that satisfy the given equation or inequality. Be sure to clearly write out all of your algebraic steps, and clearly explain the rationale behind the operations. (You may use any notation you choose to describe the set of solutions.)

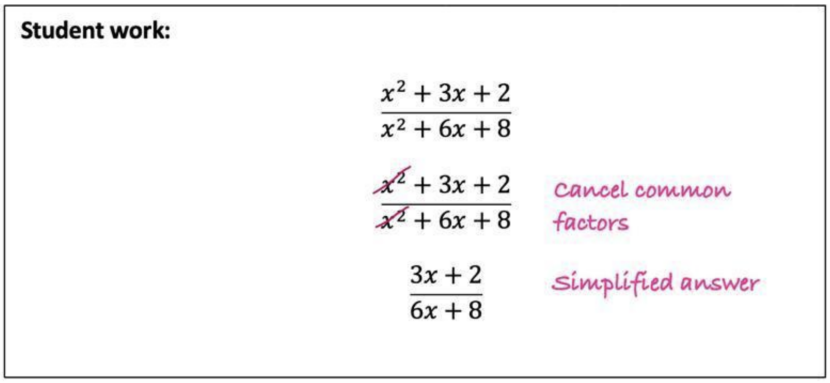
   a) $2|7+x| = \frac{1}{3}$

   b) $-(x+2)+6 < 10$

3. Simplify the following expression to eliminate the compound fraction. Be sure to clearly write out all of your algebraic steps, and clearly explain the rationale behind the operations.

$$\frac{x-1}{1-\frac{1}{x}}$$

4. **Find the Error.** A classmate simplified the rational expression $\dfrac{x^2+3x+2}{x^2+6x+8}$ in the following way:



(a) Is $\dfrac{x^2+3x+2}{x^2+6x+8}$ exactly equal to $\dfrac{3x+2}{6x+8}$ as the student claims? How can you check?

(b) What error does the student make? What are they thinking to make that error?

(c) Correctly solve the problem yourself. Clearly write your steps and explain your process in a way your classmate would understand.

Definition: For a function f defined on an interval $[a, b]$, the **average rate of change of f on** $[\mathbf{a}, \mathbf{b}]$ is

1. A cone-shaped tank is being filled with oil. Let h represent the depth of the water in the tank (measured from the bottom), and let V represent the volume of water in the tank, modeled by

$$V = f(h) = \frac{\pi h^3}{12}$$

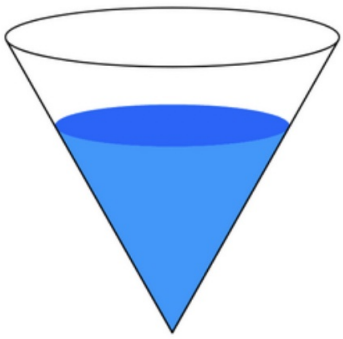
where V is measured in cubic meters and $0 \leq h \leq 12$.

(a) Find $V(1)$ and interpret the meaning in this context.

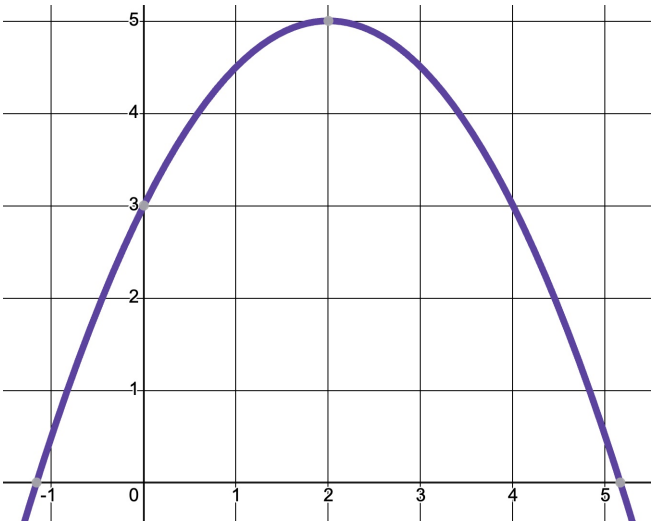
(b) Find $V(3)$ and $V(5)$. What are the units on each?

(c) Compute $AV[1, 3]$, the average rate of change from $h = 1$ to $h = 3$. What are the units? What does it mean?

(d) Which would you expect is greater? $AV[1, 3]$ or $AV[3, 5]$? Why? Compute and compare.



2. Consider the function $q(x) = 5 - \frac{1}{2}(x-2)^2$.



   (a) Estimate $AV_{[1,4]}$ using the graph.

   (b) Compute $AV_{[1,4]}$ using the equation.

   (c) A classmate has the following thinking: Since $AV_{[1,4]} = -1$ which is negative, the function is decreasing on the interval $[1, 4]$. Do you agree with the classmate? Why?

3. According to the US census, the populations of Kent and Ottawa Counties from 1970 to 2020 measured in 10-year intervals are given in the following table:

Year	**Kent**	**Ottawa**
1970	363,187	98,719
1980	411,044	128,181
1990	444,506	157,174
2000	500,631	187,768
2010	574,336	238,313
2020	602,622	263,801

   (a) Compute $AV_{[1990,2020]}$ for both Kent and Ottawa.

   (b) Explain the meaning of the average rate of change of the Ottawa county population on the time interval $[1990, 2020]$. Include units.

   (c) Which county had a greater average rate of change during the time interval $[1990, 2020]$? Were there any intervals in which one of the counties had a negative average rate of change?

F3.A Linear Functions

Definition: A line with a slope m and an initial value (vertical intercept) at $(0, b)$ has the equation in **slope-intercept form** given by:

Definition: A line with a slope m and point at (x_0, y_0) has the equation in **point-slope intercept form** given by:

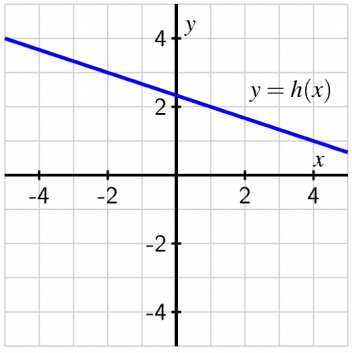
1. Find an equation for the line that is determined by the following conditions.

   (a) The line with slope $\frac{3}{7}$ that passes through $(-11, -17)$.

   (b) The line passing through the points $(-2, 5)$ and $(3, -1)$.

   (c) The line passing through $(4, 9)$ that is parallel to the line $2x - 3y = 5$.

   (d) The line in the graph:



2. The summit of Africa's largest peak, Mt. Kilimanjaro, has two main ice fields and a glacier at its peak. Geologists measured the ice cover in the year 2000 ($t = 0$) to be approximately $1951m^2$; in the year 2007, the ice cover measured $1555m^2$.

   (a) Suppose that the amount of ice cover at the peak of Mt. Kilimanjaro is changing at a constant average rate from year to year. Find a linear model $A = f(t)$ whose output is the area, A, in square meters in year t (where is the number of years after 2000).

   (b) What do the slope and y-intercept mean in the model you found in (a)? In particular, what are the units on the slope?

   (c) Compute $f(17)$. What does this quantity measure? Write a complete sentence to explain.

   (d) If the model holds further into the future, when do we predict the ice cover will vanish?

   (e) In light of your work above, what is a reasonable domain to use for the model $A = f(t)$? What is the corresponding range?

3. My used car is currently worth \$4,600. When I bought it 10 years ago, the car was already 5 years old and it cost $\$10,200$. Find a linear function $C = L(t)$ whose output is the value of the car in year t, where t is measured by the total age of the car. What is a reasonable domain for the model? What is the value and meaning of the slope of the line? Write at least one careful sentence to explain.

F3.B Quadratic Functions

1. Suppose a ball is thrown in the air. The height of the ball (in meters) is is a function of time (in seconds) given by the function

$$h(t) = -t^2 + 4t$$

   (a) What are the t intercepts? What do they represent?

   (b) What domain makes sense for this situation?

   (c) What is the vertex? What does it represent?

   (d) What does $AV_{[2,4]}$ mean? What are the units?

   (e) Find $AV_{[2,4]}$

   (f) Is there an interval where $AV_{[a,b]} = 0$?

2. Open Desmos and graph $f(x) = ax^2 + bx + c$. How do the values of a, b, and c impact the graph?

3. Reason algebraically using appropriate properties of quadratic functions to answer the following questions. Use Desmos to check your results graphically.

   (a) Find the x-intercept(s), y-intercept, and vertex of the parabola given by

   $$g(x) = 2x^2 - 2x - 40$$

   .

   (b) How many quadratic functions have x-intercepts at $(-5, 0)$ and $(10, 0)$ and a y-intercept at $(0, -1)$? Can you determine an exact formula for such a function? If yes, do so. If not, explain why.

   (c) Suppose that a quadratic function has vertex $(-3, -4)$ and opens upward. How many x-intercepts can you guarantee the function has? Why?

   (d) Does the quadratic function $w(x) = -2x^2 + 10x - 20$ have 0, 1, or 2 x-intercepts? Reason algebraically to determine the exact values of any such intercepts or explain why none exist.

F4 Composition

Definition: The function $h(t) = f(g(t))$ is called the **composition of f with g.**

The function h is defined by using the _____ of the function _____ as the _____ of the function _____.

1. Let $f(x) = x^2 - 1$, $g(x) = \dfrac{2x^2}{x-1}$, and $p(x) = \sqrt{x}$.

   (a) Find $g(f(x))$.

   (b) Find $p(g(x+h))$.

   (c) Find $f(g(p(x)))$.

2. Suppose a factory's revenue, R, is based on the gallons of paint produced, P, given by the function $R = f(P)$. Suppose also that the per hour yield of paint is a function of the number of workers, W, given by $P = g(W)$. Which of the following expressions make sense in this situation? Describe the meaning in words.

   (a) $f(g(W)) =$ (b) $g(f(W)) =$

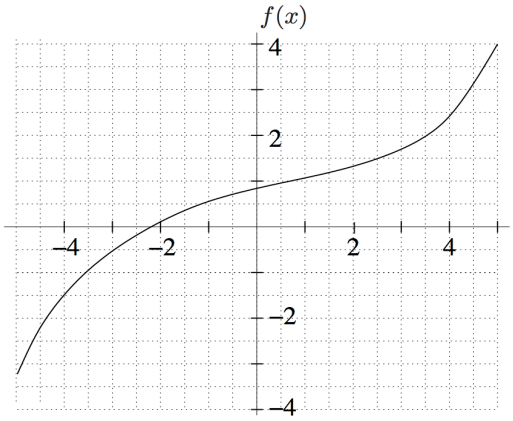
3. Use the graph f and g shown to the right to estimate each of the following.

(a) $g(f(-2)) =$

(b) $f(g(4)) =$

(c) $g(g(2)) =$

(d) $f(f(-4)) =$



x	-6	-4	-2	0	2	4	6
$g(x)$	2	0	3	7	6	1	5

4. For the function $f(x) = (x^3 + 1)^2$, find possible functions $u(x)$ and $v(x)$ such that $f(x) = u(v(x))$. Are these the only possible functions?

F5 Inverses

Definition: Suppose the functions f and g are *inverses* of each other. Then if $f(L) = m$, it must be the case that $g(\ \_\_\ ) = \_\_$.

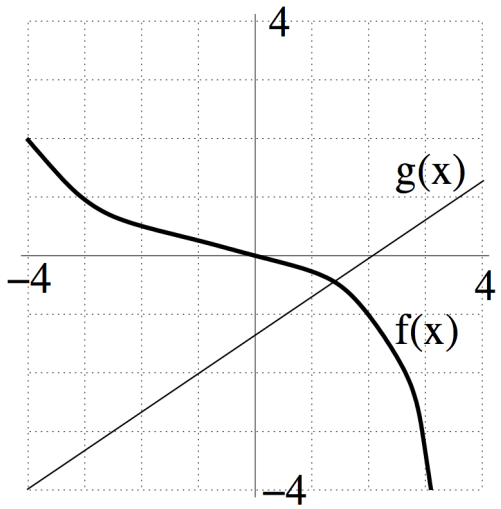
This can be written

1. Let $A = f(n)$ be the amount of periwinkle blue paint, in gallons, needed to paint n square feet of a house. Explain in practical terms what each of the following quantities represents. Use a complete sentence in each case.

   (a) $f(20)$

   (b) $f^{-1}(20)$

2. Given to the right is the graph of the functions $f(x)$ and $g(x)$. Use the function to estimate each of the following.



   (a) $f(2) =$

   (b) $f^{-1}(2) =$

   (c) $f^{-1}(g(-1)) =$

   (d) $g^{-1}(f(3)) =$

   (e) Rank the following in order from smallest to largest:
   $f(1)$, $f(-2)$, $f^{-1}(1)$, $f^{-1}(-2)$, 0.

3. Find a formula for the inverse function of each of the following functions.

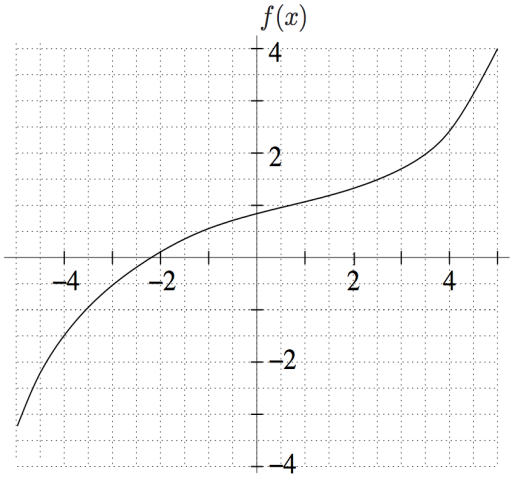
   (a) $f(x) = mx + b$

   (b) $g(x) = 2\sqrt{x-3}$

   (c) $h(x) = \dfrac{1}{3x} + 5$

   (d) $k(x) = \dfrac{3x-5}{4x+7}$

4. Use the graph and table to answer the following.



x	-6	-4	-2	0	2	4	6
$g(x)$	2	0	3	7	6	1	5

(a) $f(2) =$

(b) $f^{-1}(2) =$

(c) $g(0) =$

(d) $g^{-1}(0) =$

(e) $f(g(-4)) =$

(f) $f^{-1}(3+1) =$

(g) If $g^{-1}(t) = 0$, then $t =$

F6.A Transformations

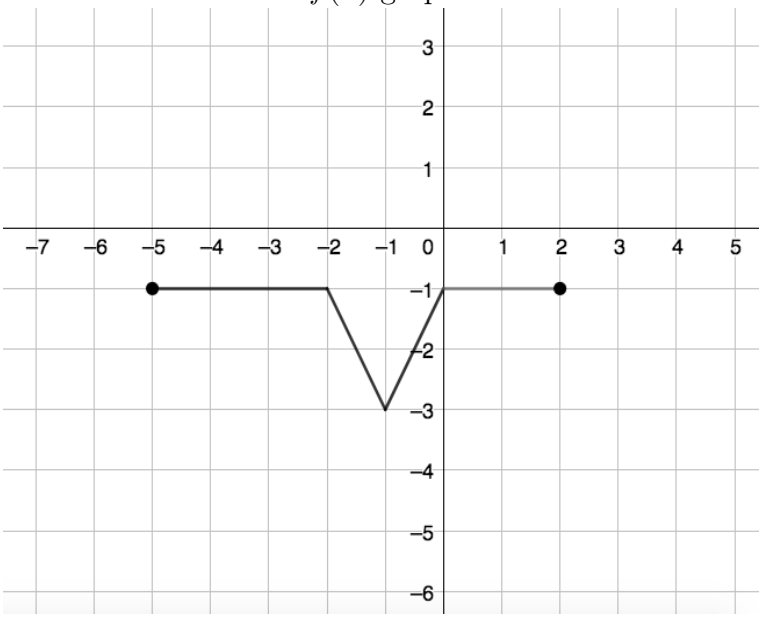
- Interpret graphically vertical and horizontal translations

Using a phone or other device, go to student.desmos.com and enter the student code. Follow the directions of the online activity.

1. Summarize your findings: How do the values of b and c impact the graph of $f(x+b)+c$? WHY?

2. Summarize your findings: How does the value of a impact the graph of $af(x)$? WHY?

3. Consider the function $f(x)$ graphed below.



Without finding an equation for the function, graph the following transformations by hand. Check your graphs using Desmos.

(a) $f(x) + 2$

(b) $f(x - 1)$

(c) $f(x + 2) + 1$

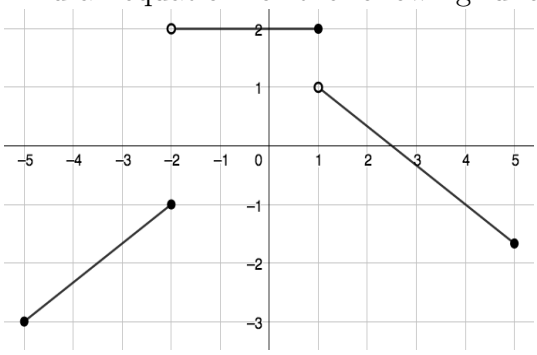
(d) $-f(x)$

(e) $2 - f(x)$

(f) $2f(x)$

F6.B Combining Functions and Piecewise Functions

1. Find an equation for the following function:



2. Let f be a function that measures a car's fuel economy in the following way. Given an input velocity v in miles per hour, $f(v)$ is the number of gallons of fuel that the car consumes per mile (i.e., "gallons per mile"). We know that $f(60) = 0.04$.

   (a) What is the meaning of the statement $f(60) = 0.04$ in the context of the problem?

   (b) Consider the function $g(v) = \dfrac{1}{f(v)}$. What is the value of $g(60)$? What are the units on g? What does g measure?

   (c) Suppose we also know that $f(70) = 0.045$. Find the average rate of change of f on the interval $[60, 70]$. What are the units on the average rate of change of f? What does this quantity measure?

3. Let $f(x) = \dfrac{|x-2|}{x-2}$.

   (a) Graph $f(x)$.

   (b) Rewrite $f(x)$ as a piecewise-defined function.

4. Last Sunday I went for a 30 minute bike ride. Over the first 5 minutes of the ride, I steadily increased my speed from 0 miles per hour to 10 miles per hour with constant rate of change. Then I held my speed at 10 mph for the next 20 minutes. During the last 5 minutes of my ride, my speed slowly decreased from 10mph to 8mph at a constant rate of change.

   (a) Let $S = f(t)$ be my speed (in mph), at time t (in minutes). Sketch a graph of the function. Be sure to clearly label each piece of my ride.

   (b) Find an equation for the piecewise function $S = f(t)$. Make sure that the input variable, t, is time in minutes from the start of the ride, and $S = f(t)$, the output, is my speed in mph.

F7 Function Review

Oregon produces nearly two thirds of the world's grass seed, with annual sales of over \$400 million. This lab models ryegrass seed production and farming using linear functions, functional representations, composition, and inverses. *Data drawn primarily from* `https://valleyfieldcrops.oregonstate.edu/grass-seed`.

1. **Overview:** In this simplified model, suppose a grass farm's revenue per acre, R, is based on the amount of ryegrass seed per acre (in pounds), G, given by the function $R = g(G)$. Suppose also that the yield of ryegrass seed is a function of the quantity of fertilizer per acre (in pounds), F, given by $G = f(F)$. Which of the following expressions make sense in this situation? Describe the meaning in words.

$$f(g(F) \qquad\qquad g(f(F))$$

2. **Fertilization:** The best fertilizers for ryegrass seed are those with low nitrogen, high phosphorus, and high potassium levels, which can have a strong impact on ryegrass seed yield per acre. Let F be the pounds of fertilizer per acre and G be the pounds of ryegrass seed yielded per acre. On average under optimal conditions, the relationship between F and G is given by the following function:

$$G = f(F) = -F^2 + 280F - 18000$$

with $120 < F < 160$.

   (a) Interpret $f(145) = 1575$. What are the units on 145? What are the units on 1575? What is the meaning in this context?

   (b) Find the vertex of the parabola generated by f. Interpret the significance of this value in this context.

   (c) Given this domain, what is the range of f? Explain.

(d) Construct a graph that displays f, including the appropriate domain and range. Label each axis. Provide a caption for your graph that explains its meaning of the important features.

3. **Fertilization Cost:** A commercial EZ-Flo fertilizer injection system has a one-time cost of \$8,389, plus an additional cost of \$1.09 per pound of fertilizer. Let C be the total cost of fertilization, and P be the total pounds of fertilizer.

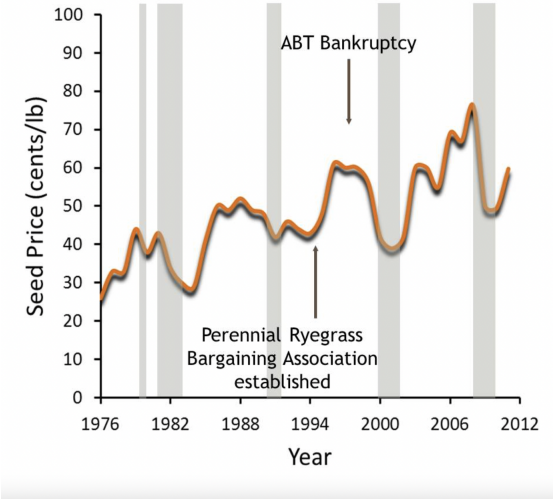
   (a) Construct a table that displays the total cost for fertilization (including the one time cost of the injection system) for the following quantities of fertilizer: $P = 1600$, $P = 8000$, $P = 16000$, $P = 32000$

   (b) Is this relationship linear? Explain.

   (c) Construct an equation for the function $C = g(P)$. Explain each component of your function. What does the vertical intercept represent in this situation? What does the value of the slope represent in this situation?

   (d) Given the results of question 1, how much would you expect a 150 acre farm to spend on fertilization? Explain your reasoning.

4. **Putting it together:** The graph below lists the ryegrass seed selling prices by year.



(a) Suppose you are the owner of a 150 acre ryegrass seed farm in 2012. How much revenue (total money) do you expect to generate? Fully explain your reasoning based on what you know about the selling price of grass seed and how much grass seed you expect to generate.

(b) Profit is revenue minus costs. If we only account for the cost of fertilizer, what equation would you use for total profit for a 150 acre farm?

(c) BONUS: How would you expect to maximize profit?

E1 Exponent Properties

Properties of Exponents:

1. $a^n a^m =$_____
2. $\dfrac{a^n}{a^m} =$_____
3. $(a^n)^m =$_____
4. $(ab)^n =$_____
5. $\left(\dfrac{a}{b}\right)^n =$_____

Some Definitions:

1. $a^0 =$_____
2. $a^{-n} =$_____
3. $a^{\frac{1}{n}} =$_____
4. $a^{\frac{m}{n}} =$_____

1. Simplify the following:

   (a) $a^3 \cdot a^7$

   (b) $\left(3m^4\right)^2$

   (c) $\left(3 + k^4\right)^2$

   (d) $\left(\dfrac{2x}{5}\right)^3$

   (e) $\dfrac{n^{-1}a}{a^2}$

   (f) $\dfrac{s^{-1}t + 1}{s^2}$

2. Rewrite the following to include no radicals or negative exponents.

(a) $\dfrac{(5xy^3)^2}{x^0y^5}$

(b) $\dfrac{(AB)^4}{A^{-1}B^{-2}}$

(c) $\left(\dfrac{a^3b^{-1}}{\sqrt{a^{\frac{5}{2}}}}\right)^{-2}$

(d) $2b^{-1}(b^2+b)-2$

(e) $\sqrt{x^e y^{\frac{e}{2}}} + (x^e)(x^e)^2$

(f) $\sqrt[3]{t^3 + 7(t^9)^{\frac{1}{3}}}$

(g) Solve for x: $81^x = 3$

(h) Simplify $9^{-\frac{1}{2}} + \sqrt{0.01}$ without a calculator.

E2.A Exponential Growth and Decay

- Modeling situations with exponential functions
- Shapes of exponential functions
- Finding equations of exponential functions

Warm-up: Simplify the following exponential expressions:

1. $\sqrt{9+x^2}$
2. $\dfrac{4a^3}{a^{-2}}$
3. $s^3 - s^2$

Definition: An exponential function $Q = f(t)$ has the formula $f(t) = ab^t$, where $b > 0$.

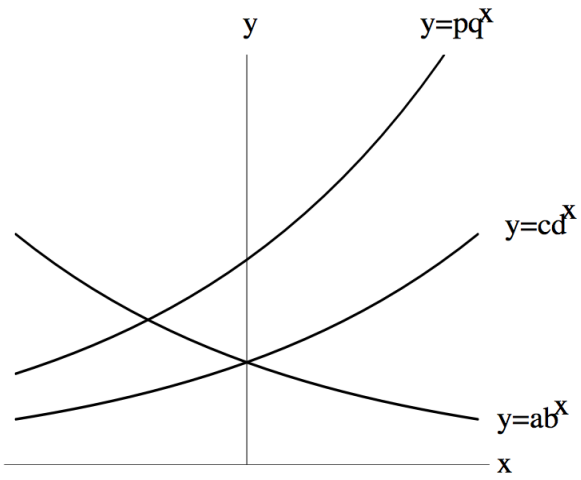
a is _____

b is _____

And $b = 1 + r$ where r is _____

1. The population of California Sea Lions, P, is currently 300,000 and decreasing at a rate of 5% per year. Find an equation that models the situation and sketch the graph.

2. Consider the exponential graphs pictured below and the six constants a, b, c, d, p, and q.



(a) Place in order from least to greatest: 0, p, c, a.

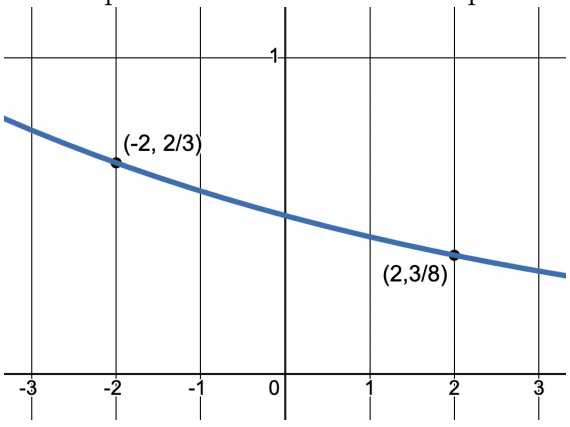
(b) Place in order from least to greatest: -1, 0, 1, q, d, b.

(c) Which two of these constants could be equal?

3. The table below gives incomplete data for the temperature of a substance during an experiment. Construct an exponential equation for the relationship between the temperature an time and complete the missing elements of the table.

t (minutes)	0	5	10	15	20
T (°C)		19.33	31.12		80.73

4. Find a possible formula for the exponential function shown below.



E2.B Modeling with Exponential Functions

- Modeling situations with exponential functions
- Shapes of exponential functions

Warm-up: Compare the following expressions. Which are equivalent?

1. $x^{\frac{1}{2}}$
2. $\sqrt{x}$
3. x^{-2}

1. Graph $f(t) = ab^t + c$ in Desmos. Describe how each value a, b, and c impact the shape of the graph?

2. Let $g(t) = 4(2^t) - 6$. Without using Desmos, describe the important features of the graph:

   (a) Describe the end behavior of g.

   (b) What are the vertical and horizontal intercepts of g?

   (c) Is g concave up or concave down?

   (d) Check you thinking using Desmos.

3. A can of soda (at room temperature) is placed in a refrigerator at time t (in minutes) and its temperature, T, in degrees Fahrenheit, is computed at regular intervals. Based on the data, a model is formulated for the object's temperature, given by

$$F(t) = 42 + 30(0.95)^t$$

   (a) Consider the simpler function $p(t) = 30(0.95)^t$. Describe the behavior of this function and sketch a graph.

   (b) Consider the full function $p(t) = 42 + 30(0.95)^t$. Describe the behavior of this function and sketch a graph.

   (c) What is the temperature of the refrigerator? What is the room temperature of the surroundings outside the refrigerator? Why?

   (d) Determine the average rate of change of F on the intervals $[10, 20]$, $[20, 30]$, and $[30, 40]$. Write at least two careful sentences that explain the meaning of the values you found, including units, and discuss any overall trend in how the average rate of change is changing.

4. A researcher is investigating the impact of a ecological recovery project on the population of Long Leaf Pine trees at two different site locations, where number of pines are given as trees per acre. (This problem will require the use of a graphing calculator.)

Year	1995	2000	2005	2010	2015
Location A	456	485	520	556	591
Location B	132	159	193	234	283

   (a) Find two formulas: one for the density of Long Leaf Pines at Location A and the other at Location B.

   (b) Use your formulas to predict the density of Long Leaf Pines at each location in 2022.

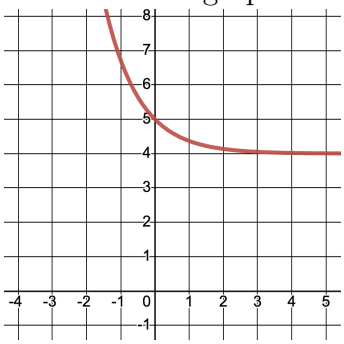
   (c) Will the density of trees ever be the same at both locations? If so, when? Explain your prediction.

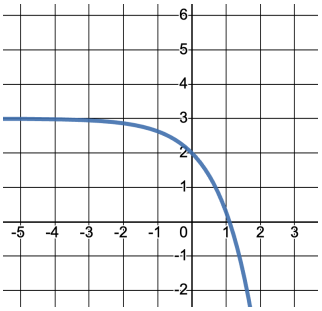
E2.C The Number e

Warm-up:
Simplify $\sqrt{n}\left(n^{\frac{7}{2}}+n^3\right)$

Definition: The number e is named after mathematician Leonhard Euler. It is an irrational number (continues without repeating), approximated by 2.7182818284590452 35...

1. The functions graphed below are translations of the graph $f(x)=e^x$.





   (a) What is the horizontal asymptote of each function?

   (b) Find equations for the functions.

2. Use Desmos to visually estimate the solutions to the following equations.

   (a) $e^t = 2$

   (b) $2e^t - 4 = 7$

   (c) $4 - 2e^{-0.7t} = 3$

   (d) $2e^{1.2t} = 1.5e^{1.6t}$

3. A group of researchers are conducting an experiment that includes the measurement of a population of insects over a short period of time. The population of crane flies is given as $P = 45e^{0.9t}$, where t is measured in days after the start of the experiment.

   (a) What is the initial population of crane flies? How many flies are there after 1 day?

   (b) By what percentage does the population of flies grow each day?

   (c) Rewrite the equation for the number of flies in the form $F = a \cdot b^t$. What do F, P, a and b represent?

   (d) Use a graphing utility to estimate when the population of flies reaches 5,000.

4. BONUS: Estimate the value of $\left(1 + \frac{1}{n}\right)^n$ as n gets large.

E3.A What is a Logarithm?

Definition Let x be a positive number.

1. If $\log_a x = y$ then ...

2. $\log x$ is ...

3. $\ln x$ is ...

1. Calculate each of the following, exactly if possible.

   (a) $\log_2 1 =$

   (b) $\log 100 =$

   (c) $\ln e =$

   (d) $\log_5 5 =$

   (e) Bonus: $\log_{16} 8 =$

Facts about Logarithms

1. (a) $\log_a(a^x) =$

   (b) $a^{\log_a x} =$

2. (a) $\log_a 1 =$

   (b) $\log_a a =$

2. What is the domain and what is the range of $f(t) = \ln(t)$. Why?

3. Solve each equation for t. Leave as an exact value.

(a) $e^t = \dfrac{1}{10}$

(b) $5e^t = 7$

(c) $\ln(t) = -\dfrac{1}{3}$

(d) $2^{1-3t} = 4$

(e) $2\log(t) + 1 = 4$

(f) $4 - 3e^{2t} = 2$

(g) $5^{2t} = 5^{5t+1}$

(h) $\ln(5 - 6t) = -2$

E3.B Properties of Logs

Facts about Logarithms

1. (a) $\log_a(a^x) =$

   (b) $a^{\log_a x} =$

2. (a) $\log_a 1 =$

   (b) $\log_a a =$

3. (a) $\log_a(nm) =$

   (b) $\log_a\left(\frac{n}{m}\right) =$

   (c) $\log_a(n^t) =$

Change of base formula:

$$\log_b(a) = \frac{\ln(a)}{\ln(b)}$$

1. Solve each equation for x. Leave as an exact value.

   (a) $5 \cdot 4^x = 25$

   (b) $5x^4 = 25$

   (c) $5x^9 = 10$

   (d) $\log(x^2) = -2$

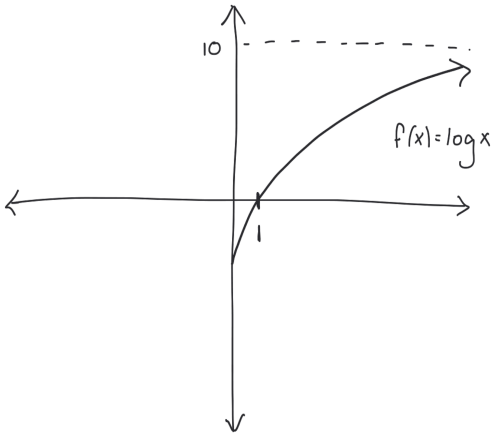
   (e) $10e^{4x+1} = 20$

   (f) $\ln(x+5) = 10$

   (g) $5^{2x} + 5^{2x} = 1$

   (h) $5 \cdot 3^x = 2 \cdot 7^x$

2. A classmate graphed the function $f(x) = \log(x)$, shown below. How well does this graph capture the important features of the function? What looks good? What could be improved? Sketch your own graph.



3. Decide whether each statement is true for all values of x and y. If false, find a way to correct the statement.

   (a) $\ln(x + y) = \ln x + \ln y$

   (b) $\ln(xy^2) = \ln x + 2 \ln y$

   (c) $\log(1/x) = -\log x$

E3.C Modeling with Logs

Warm-up:
Solve for t:

$$\log(3t) + \log(t) = 6$$

1. A can of soda is initially at room temperature, 72.3° Fahrenheit, and at time $t = 0$ is placed in a refrigerator set at 37.7°. In addition, we know that after 30 minutes, the soda's temperature has dropped to 59.5°.

   (a) Use algebraic reasoning and your understanding of the physical situation to determine the exact values of a, c, and k in the model $F(t) = ae^{-kt} + c$. Write at least one careful sentence to explain your thinking.

   (b) Determine the exact time the object's temperature is 42.4°. Clearly show your algebraic work and thinking.

   (c) Find the average rate of change of on the interval $[20, 30]$. What is the meaning (with units) of this value?

   (d) If everything stayed the same except the value of $F(0)$, and instead $F(0) = 65$, would the value of k be larger or smaller? Why?